Correlation and Regression

Correlation and regression are the two most commonly used techniques for investigating the relationship between quantitative variables. Here regression refers to linear regression. Correlation is used to give the relationship between the variables whereas linear regression uses an equation to express this relationship.

Correlation and regression are used to define some form of association between quantitative variables that are assumed to have a linear relationship. In this article, we will learn more about these topics, the difference between correlation and regression as well as see some associated examples.

What are Correlation and Regression?

Correlation and regression are statistical measurements that are used to give a relationship between two variables. For example, suppose a person is driving an expensive car then it is assumed that she must be financially well. To numerically quantify this association, correlation and regression are used.

Correlation Definition

Correlation can be defined as a measurement that is used to quantify the relationship between variables. If an increase (or decrease) in one variable causes a corresponding increase (or decrease) in another then the two variables are said to be directly correlated. Similarly, if an increase in one causes a decrease in another or vice versa, then the variables are said to be indirectly correlated. If a change in an independent variable does not cause a change in the dependent variable then they are uncorrelated. Thus, correlation can be positive (direct correlation), negative (indirect correlation), or zero. This relationship is given by the correlation coefficient.

Regression Definition

Regression can be defined as a measurement that is used to quantify how the change in one variable will affect another variable. Regression is used to find the cause and effect between two variables. Linear regression is the most commonly used type of regression because it is easier to analyze as compared to the rest. Linear regression is used to find the line that is the best fit to establish a relationship between variables.

Correlation and Regression Analysis

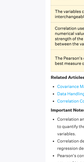
Both correlation and regression analysis are done to quantify the strength of the relationship between two variables by using numbers. Graphically, correlation and regression analysis can be visualized using scatter plots.

Correlation analysis is done so as to determine whether there is a relationship between the variables that are being tested. Furthermore, a correlation coefficient such as Pearson's correlation coefficient is used to give a signed numeric value that depicts the strength as well as the direction of the correlation. The scatter plot shows the correlation between two variables x and y for individual data points as shown below.

Correlation Analysis Graph

Regression analysis is used to determine the relationship between two variables such that the value of the unknown variable can be estimated using the knowledge of the known variables. The goal of linear regression is to find the best-fitted line through the data points. For two variables, x_i and y_i the regression analysis can be visualized as follows:

Regression Analysis Graph



Correlation and Regression Formula

The best way to conduct correlation and regression analysis is by using Pearson's correlation coefficient and by adopting the method of least squares respectively. The correlation and regression formula is given below:

Pearson's Correlation Coefficient: $r_{xy} = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x})^2 \Sigma(y_i-\bar{y})^2}}$

Ordinary Least Squares (OLS) Linear Regression:

The straight line equation is given as $y = \alpha + \beta x$

$$\beta = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2}$$

$$\beta = r_{xy}\frac{\sigma_y}{\sigma_x}$$

$$\alpha = \bar{y} - \beta\bar{x}$$

Here, $\bar{x}$ is the mean, and σ_x is the standard deviation of the first data set where each data point is represented by x_i. Similarly, $\bar{y}$ is the mean, and σ_y is the standard deviation of the second data set. n is the number of data points in the datasets.

Difference between Correlation and Regression

Correlation and regression are both used as statistical measurements to get a good understanding of the relationship between variables. If the correlation coefficient is negative (or positive) then the slope of the regression line will also be negative (or positive). The table given below highlights the key difference between correlation and regression.

Correlation	Regression
Correlation is used to determine whether variables are related or not.	Regression is used to numerically describe how a dependent variable changes with a change in an independent variable
Correlation tries to establish a linear relationship between variables.	It finds the best-fitted regression line to predict the value of one variable on the basis of the known variable.
The variables can be used interchangeably	The variables cannot be interchanged.
Correlation uses a signed numerical value to estimate the strength of the relationship between the variables.	Regression is used to show the impact of a unit change in the independent variable on the dependent variable.
The Pearson's coefficient is the best measure of correlation.	The least-squares method is the best technique to determine the regression line.

Related Articles:

- Covariance Matrix
- Data Handling
- Correlation Coefficient Calculator

Important Notes on Correlation and Regression

- Correlation and regression are statistical measurements that are used to quantify the strength of the linear relationship between two variables.
- Correlation determines if two variables have a linear relationship while regression describes the cause and effect between the two.
- Pearson's correlation coefficient and ordinary least squares method are used to perform correlation and regression analysis.

Examples on Correlation and Regression

Example 1: Calculate the correlation coefficient for the given data

Person	Hand	Height
A	17	150
B	15	154
C	19	169
D	17	172
E	21	175

Solution:

Person	Hand (x_i)	Height (y_i)	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
A	17	150	-0.8	-14.0	11.2	0.6	196.0
B	15	154	-2.8	-10.0	28.0	7.8	100.0
C	19	169	1.2	5.0	6.0	1.4	25.0
D	17	172	-0.8	8.0	-6.4	0.6	64.0
E	21	175	3.2	11.0	35.2	10.2	121.0
Average	17.8	164		Total	74.0	20.8	506.0

Using the formula,

$$r_{xy} = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x})^2 \Sigma(y_i-\bar{y})^2}}$$

$$= 0.72$$

Answer: The data has a high positive correlation

Example 2: Find the equation of the regression line for the following data that has a correlation coefficient equal to 0.61

Person	Weight	Blood Pressure
A	150	125
B	169	130
C	175	160
D	180	169
E	200	190

Solution:

Person	Weight	Blood Pressure	$(x_i - \bar{x})$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^2$
A	150	125	-24.8	-21.8	615.0	475.2
B	169	130	-5.8	-16.8	33.6	282.2
C	175	160	0.2	13.2	0.0	174.2
D	180	169	5.2	22.2	27.0	492.8
E	200	190	25.2	3.2	635.0	10.2
Average	174.8	146.8		Total	1310.8	1434.6

$$\sigma_x = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n}} = 18.3$$

$$\sigma_y = \sqrt{\frac{\Sigma(y_i-\bar{y})^2}{n}} = 18.94$$

$$r_{xy} = 0.61$$

$$\beta = r_{xy}\frac{\sigma_y}{\sigma_x} = 0.64$$

$$\alpha = \bar{y} - \beta\bar{x} = 146.8 - (174.8)(0.64) = 35.21$$

Equation of regression line: $y = 35.21 + 0.64x$

Answer: $y = 35.21 + 0.64x$

Example 3: Interpret the correlation coefficient of the following data

X	Y
12	1002
14	780
21	580
34	400
56	350
78	120

Solution:

X	Y	$(x_i - \bar{x})$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
12	1002	-23.833	466.67	-11122.22	568.028	21777.78
14	760	-21.833	224.67	-490 5.22	476.804	5047 5.11
21	580	-14.833	44.67	-662.56	220.028	1995.11
34	400	-1.833	-135.33	248.11	3.361	18315.11
56	350	20.167	-185.33	-3737.56	406.694	3434 8.44
78	120	42.167	-415.33	-17511 3.22	1778.028	1725 0.78
$\bar{x} = 35.83$	$\bar{y} = 535.83$		Total	-27692.67	3452.833	4954 13.33

Using the formula,

$$r_{xy} = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x})^2 \Sigma(y_i-\bar{y})^2}}$$

$$= -0.914$$

Answer: There is a strong negative correlation between the two variables

FAQs on Correlation and Regression

What are Correlation and Regression in Statistics?

In statistics, correlation and regression are measures that help to describe and quantify the relationship between two variables using a signed number.

What is the Definition of Correlation and Regression?

Correlation in correlation and regression can be defined as a numeric value that determines whether variables are linearly related and give a numeric value to the corresponding strength. Regression is an equation that checks how a change in one variable will result in a change in another variable.

What is the Formula for Correlation and Regression?

The formula for correlation and regression is given as follows

- Correlation: $r_{xy} = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x})^2 \Sigma(y_i-\bar{y})^2}}$
- Regression line equation: $y = \alpha + \beta x$, where $\beta = r_{xy}\frac{\sigma_y}{\sigma_x}$ and $\alpha = \bar{y} - \beta\bar{x}$.

What is the Similarity Between Correlation and Regression?

The similarity between correlation and regression is that if the correlation coefficient is positive (or negative) then the slope of the regression line will also be positive (or negative).

What is the Difference Between Correlation and Regression?

The main difference between correlation and regression is that correlation is used to find whether the given variables follow a linear relationship or not. Regression is used to find the effect of an independent variable on a dependent variable by determining the equation of the best-fitted line.

How to Graphically Represent Correlation and Regression?

A scatter plot or scatter chart is used to represent correlation and regression graphically. The data points of the variables are plotted on the graph to check the correlation and the best-fitted line represents the regression equation.

What is the Best Way to Find Correlation and Regression Between Two Variables?

The best way to find the correlation and regression between two variables is by using Pearson's correlation coefficient and by employing the ordinary least squares method respectively.